Week 9 Interim Submission

Designing and Prototyping a Human-Centred Clinical System (HCI/HRI)

Student: Vishal Baboolal

Project: AI-Assisted Emergency Department Triage

Preferred implementation: HCI nurse-facing ED triage desk

Interim scope completed: HCI problem-space canvas, HRI problem-space canvas, preferred HCI co-design canvas, initial HCI/HRI mock-ups, system requirement notes, comparison and safety considerations.

The final video and polished interactive prototype are not required for the interim deadline.

1. Design direction and rationale

The preferred implementation is a screen-based HCI tool at the main ED triage desk. It places the current model in a context where the primary user is a trained nurse, the output can be reviewed quickly, and failures can fall back to the existing manual triage protocol. The Observation Unit HRI concept is retained as a comparison and future design direction, but it introduces additional physical, sensor, trust and proximity risks.

Interim deliverables included

- Problem-space co-design canvas for the HCI ED triage desk.
- Problem-space co-design canvas for the HRI Observation Unit robotic kiosk.
- Expanded preferred HCI canvas covering Problem, Ethics, Guidelines, MVP, Environment and Form.
- Initial low-fidelity mock-ups for both settings.
- System-requirement notes covering inputs, outputs, human action, integration, failure modes and safety.
- Three HCI-specific and three HRI-specific safety considerations in Concern-Context-Mitigation-Residual Risk format.

2. HCI Problem-Space Canvas

HCI - ED Triage Desk

Problem Space Co-Design Canvas - Week 9 Interim

User group(s)

- Primary: ED triage nurses
- Secondary: charge nurse, ED physicians, clinical IT, quality/governance staff
- Indirect: patients whose care is prioritised using the nurse's final decision

Characteristics

- Clinically trained; works under time pressure and frequent interruptions
- May be fatigued during 12-hour shifts
- Needs rapid, glanceable information and a reliable manual fallback

Needs

- A recommendation that is visible in seconds
- Clear missing-data and uncertainty warnings
- A fast confirm/override workflow with accountability
- No replacement of professional judgement

User goals

Short term

- Reduce time spent assembling pre-triage data
- Make high-acuity patterns easier to notice
- Standardise documentation of confirm/override decisions

Long term

- Support a safe silent pilot
- Improve consistency without deskilling nurses
- Create auditable evidence for local validation and future refinement

AI SYSTEM task(s)

Short term

- Pull or accept patient data
- Validate inputs and identify missing values
- Display proposed ESI level 1-5 with text, icon, colour and confidence
- Record nurse confirmation or override reason

Long term

- Summarise aggregate use and override patterns
- Support review of performance, fairness, drift and workflow fit
- Never auto-finalise a clinical decision

Advantages / why this setting may benefit

- **Fast visual scanning**
- **No physical movement near patients**
- **Works with trained staff**
- **Easy to log and review**
- **Supports manual fallback**

3. HRI Problem-Space Canvas

HRI - Observation Unit Robotic Kiosk

Problem Space Co-Design Canvas - Week 9 Interim

User group(s)

- Primary: Observation Unit nurses
- Secondary: patients, porters, biomedical/IT technicians
- Oversight: physicians, Infection Prevention, governance and facilities teams

Characteristics

- Mixed expertise and comfort with robots
- Patients may be anxious, unwell, multilingual or physically limited
- Staff work in a noisy shared environment with limited space

Needs

- A stationary, predictable and non-threatening system
- Reliable vitals capture with clear sensor-failure handling
- Nurse control over every classification and escalation
- Easy disengagement, privacy and physical clearance

User goals

Short term

- Capture selected vital signs consistently
- Route a clear summary to the nurse
- Reduce repeated manual transcription where safe

Long term

- Assess whether robot-assisted intake adds value without distress
- Build trust through transparency and safe degradation
- Determine whether future physical integration is justified

ROBOTIC SYSTEM task(s)

Short term

- Guide patient positioning for approved sensors
- Stream device readings and flag invalid/missing signals
- Display staff-facing recommendation only after nurse review
- Stop safely and request manual assessment when uncertain

Long term

- Remain stationary during pilot
- Log sensor failures, overrides and interruptions
- Avoid autonomous movement, diagnosis or treatment

Advantages / why this setting may benefit

- **Direct device data capture**
- **Consistent prompts**
- **Physical presence can guide positioning**
- **Can operate beside a vitals station**
- **Creates clear opportunities for fail-safe design**

4. Preferred HCI Canvas - Ethics

Preferred HCI Implementation - Ethical Considerations

ED triage desk, nurse-facing decision support

Physical safety

Problem

A delayed or confusing alert may contribute to delayed escalation.

Solution

Critical red flags remain visible; manual triage is always available; the interface never blocks care.

Data security

Problem

Patient details could be exposed through weak access control or screen visibility.

Solution

Role-based login, automatic timeout, encryption, audit logging and no patient data in browser logs.

Transparency

Problem

A numeric ESI output can appear more certain or authoritative than it is.

Solution

Show confidence status, missing inputs, data source and a clear 'nurse judgement is final' statement.

Equality across users

Problem

Model performance and interface access may vary by subgroup or disability.

Solution

No colour-only meaning; accessible interaction; subgroup monitoring and local fairness review before expansion.

Emotional consideration

Problem

An urgent alert can increase stress or imply that the nurse made an error.

Solution

Neutral, respectful wording; no blame language; support escalation without shaming the user.

Behaviour enforcement

Problem

Automation bias may pressure nurses to accept the model recommendation.

Solution

Confirm and override controls are equally visible; override requires a reason for audit, not punishment.

5. Preferred HCI Canvas - Guidelines

Preferred HCI Implementation - Design Guidelines

Guidelines that govern the ED triage desk interface

Environment guidelines

- Optimise for a noisy, bright, cluttered and interruption-heavy ED.
- Assume the user may be fatigued late in a 12-hour shift.
- Maintain a complete manual pathway during power or connectivity loss.
- Fit the existing desk without adding new physical obstruction.

Form guidelines

- Use high contrast, large type and large touch targets.
- Keep the proposed ESI level and missing-data state above the fold.
- Use text, icon and shape in addition to colour.
- One primary action per screen; avoid dense dashboards.

Interaction guidelines

- Show the recommendation only after required inputs are validated.
- Provide confirm and override from the same screen.
- Make manual mode reachable within two actions.
- Do not use routine audio; never hide system status.

Behaviour guidelines

- Use neutral, non-authoritative language.
- Never auto-finalise ESI or initiate treatment.
- When uncertain, ask for manual review rather than guessing.
- Log failures and overrides without penalising the nurse.

6. Preferred HCI Canvas - MVP

Preferred HCI Implementation - Minimum Viable Product

The smallest pilot that delivers value without replacing clinical judgement

Where and when

- Main ED triage desk
- Silent pilot during staffed shifts
- Nurse-facing only

System role

- Summarise valid inputs
- Propose ESI 1-5
- Flag uncertainty and missing data

Interaction tone

- Calm, concise and professional
- Supportive, never commanding
- Plain language

MVP features

- EHR pull with manual fallback
- Vital-sign and chief-complaint summary
- Proposed ESI with text + icon + colour
- Confidence / missing-data warning
- Confirm and override controls
- Audit log and final nurse decision

Connection to systems

- Read-only Mercer EHR interface
- Hospital authentication
- Audit-log storage
- Offline/manual mode
- No treatment orders or autonomous workflow actions

Pilot success measures

- Median response time below 1.5 seconds
- Manual fallback completed without confusion
- Nurses can identify recommendation and missing-data state within 3 seconds
- Override events are completely logged
- No adverse event attributable to the interface during the silent pilot

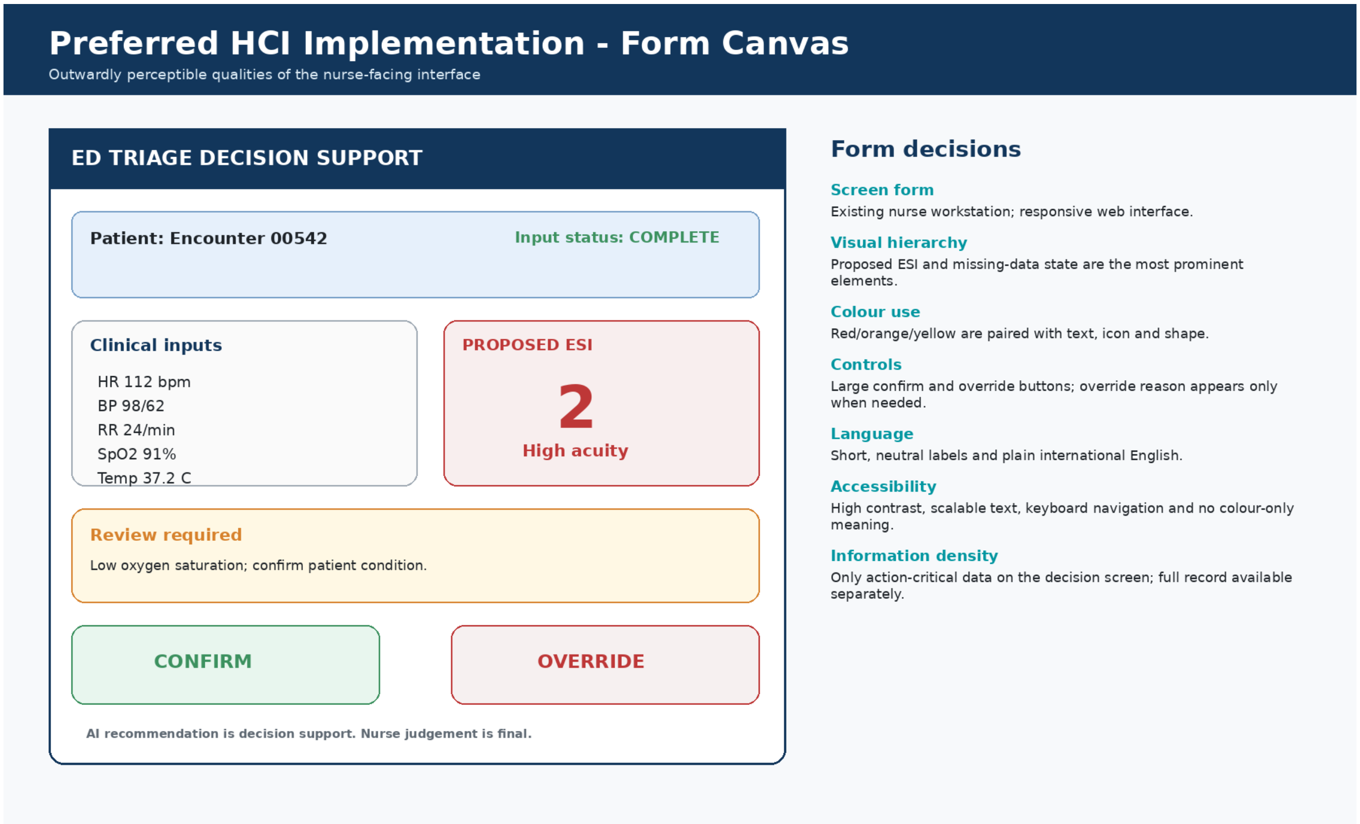
7. Preferred HCI Canvas - Environment

Preferred HCI Implementation - Environment Canvas

Mercer operating conditions and the design response

	Operating condition	Design response
Noise	Ambient noise may exceed 70 dB; audio competes with existing alarms.	Use visual-first alerts; reserve audio for approved exceptions only.
Lighting	Fluorescent light, shadows and changing ambient conditions affect legibility.	High contrast, large type, matte display and no colour-only meaning.
Shift fatigue	Nurses may work 12-hour shifts and make decisions under interruption.	Short labels, one primary action, predictable screen order and visible system state.
Power instability	Generator switching and brief outages are foreseeable.	UPS-compatible workstation, safe recovery, offline banner and manual protocol.
Physical space	The desk is small, shared and cluttered.	Use existing screen; no extra hardware in the nurse's movement path.
Privacy	Patient details may be visible to people near the desk.	Role-based access, automatic timeout, privacy-aware layout and minimal on-screen identifiers.

8. Preferred HCI Canvas - Form



9. HCI versus HRI implementation comparison

Dimension	Setting A - ED triage desk (HCI)	Setting B - Observation Unit (HRI)
Primary user	ED triage nurse	Observation nurse; patient participates in vitals capture
Primary input route	EHR pull; manual entry fallback	Device stream; nurse verification/manual fallback
Model output	ESI 1-5, colour + text + icon, confidence, missing-data warning	Staff-facing display and status light; no autonomous diagnosis
Human next step	Review, confirm or override; document reason	Verify vitals, accept/manual-assess, or stop interaction
Latency tolerance	Recommendation within 1.5 seconds	Device acquisition may take longer, but time-critical red flags bypass the robot
Failure consequence	Nurse returns to manual triage protocol	Robot freezes safely, becomes non-interactive and alerts staff
Preferred interim direction	YES - safer and more feasible for current pilot	Comparison concept only for this phase

Decision: Proceed with the HCI nurse-facing screen for the current prototype. Use the HRI kiosk only as a comparison concept until physical and sensor risks can be tested.

10. System vision and workflow

The system is a nurse-facing clinical decision-support interface that receives pre-triage data, validates the inputs, runs the pinned model and presents a proposed ESI level. The nurse reviews the output, confirms or overrides it, and remains responsible for the final clinical decision. The tool is intended for offline research and a silent pilot; it must not make treatment decisions or replace the manual triage process.

Users and stakeholders

- Primary user: ED triage nurse.
- Secondary clinical users: charge nurse and ED physicians.
- Operational stakeholders: Clinical IT, cybersecurity, quality/governance and hospital administration.
- Indirect stakeholders: patients and caregivers affected by workflow and prioritisation.

Inputs, outputs and required human action

Element	HCI ED desk	HRI Observation Unit	Fallback / human action
Input route	EHR pull; manual entry fallback	Device stream; nurse verification/manual fallback	Use manual protocol when source is unavailable
Data consumed	Vital signs, chief complaint, approved pre-triage fields	Approved vitals sensors plus staff-entered context	Do not infer from partial required data
Output	ESI 1-5, text/icon/colour, confidence, missing-data state	Staff-facing display/status; no autonomous diagnosis	Nurse receives a clear manual-review state
Human action	Confirm or override and document final ESI	Verify vitals, accept/manual-assess or stop interaction	Nurse remains final decision-maker
No action	Recommendation remains pending; no auto-finalisation	Robot stays stationary and non-interactive	Escalation follows approved clinical protocol, not autonomous AI action

11. Functional requirements

ID	Testable requirement
FR-01	The system shall require staff authentication before showing identifiable patient information.
FR-02	The system shall accept an EHR data pull as the primary route and manual entry as the fallback route.
FR-03	The system shall validate required fields and clearly identify missing, stale or implausible inputs before inference.
FR-04	The system shall display the proposed ESI classification within 1.5 seconds of valid submission.
FR-05	The output shall include ESI level 1-5, a text label, an icon, a colour state, confidence status and the input values used.
FR-06	The interface shall never use colour as the only indicator of urgency.
FR-07	The nurse shall be able to confirm or override the recommendation from the same screen.
FR-08	An override shall require a selectable reason and optional free-text note.
FR-09	Low confidence, missing required data or system failure shall route the nurse to the manual triage protocol.
FR-10	The system shall log the recommendation, final nurse decision, override reason, timestamp and user ID.
FR-11	The system shall not auto-finalise an ESI level or initiate treatment.
FR-12	The system shall provide a persistent statement that the AI is decision support and nurse judgement is final.

12. Non-functional requirements

ID	Testable requirement
NFR-01	The interface shall remain legible under fluorescent lighting, shadows and variable ambient conditions.
NFR-02	A trained nurse shall be able to identify the proposed ESI level and missing-data state within 3 seconds.
NFR-03	Touch targets shall be large enough for hurried use and the workflow shall also support keyboard navigation.
NFR-04	Critical meaning shall use text, icon and shape in addition to colour.
NFR-05	The system shall degrade gracefully during power, device or connectivity interruption.
NFR-06	The system shall avoid routine audio alerts; any audio shall be reserved for approved high-priority states.
NFR-07	All patient data shall be encrypted in transit and access shall be role based.
NFR-08	The interface shall avoid exposing patient data in browser logs, screenshots or unauthorised displays.

13. Integration requirements

ID	Integration requirement
IR-01	The system shall connect to the Mercer EHR through a read-only, approved interface such as FHIR R4 or a Mercer-approved API.
IR-02	If the EHR connection is unavailable, manual entry shall be reachable within two user actions.
IR-03	The system shall record source of information for each input: EHR, manual or device.
IR-04	The system clock shall synchronise with the hospital time source for audit logging.
IR-05	The deployment shall use hospital-approved authentication, encryption and audit-log storage.
IR-06	The workstation shall remain usable on generator/UPS power and visibly indicate offline mode.
IR-07	The Observation Unit concept shall accept device data only from approved, validated sensors and shall support immediate manual fallback.

14. HCI-specific safety considerations

1. Alarm fatigue

Concern	Excessive alerts cause nurses to ignore or silence the system.
Context	A busy ED already contains many alarms and interruptions, especially during long shifts.
Mitigation	No routine audio; persistent visual alert only for approved high-acuity states; suppress duplicates; review alert rate during pilot.
Residual risk	Thresholds may still be miscalibrated for Mercer and must be reviewed during local testing.

2. Display legibility under stress

Concern	Critical information becomes hard to read under variable lighting, clutter and divided attention.
Context	The triage desk is small, shared and visually busy; the nurse may glance for less than three seconds.
Mitigation	High contrast, large type, one primary action, uncluttered layout, clear missing-data state and short labels.
Residual risk	Individual differences and unusual lighting may still affect readability; user testing is required.

3. Colour blindness and accessibility

Concern	Colour-only urgency states can exclude users or cause misinterpretation.
Context	The interface may be used by staff with varied visual ability, language background and device settings.
Mitigation	Pair colour with text, icon and shape; keyboard access; plain language; accessible contrast and scalable text.
Residual risk	Not all needs are known until tested with real staff and accessibility review.

15. HRI-specific safety considerations

1. Voice input failure in noise

Concern	The robot mishears patient or staff speech and records incorrect information.
Context	ED/Observation ambient noise, accents, masks and multilingual speech reduce recognition accuracy.
Mitigation	Voice is supplementary only; all captured responses require visible confirmation; touch/manual entry always available.
Residual risk	Voice may still be unusable for some users, so the workflow must remain complete without it.

2. Sensor failure and graceful degradation

Concern	A failed sensor creates a misleading classification or abandons the assessment.
Context	Device dropout and power instability are foreseeable in the deployment environment.
Mitigation	Range and signal-quality checks; stop inference when required data fails; alert nurse; log fault; immediate manual fallback.
Residual risk	A busy nurse may be delayed in responding, so no classification is safer than a partial-data classification.

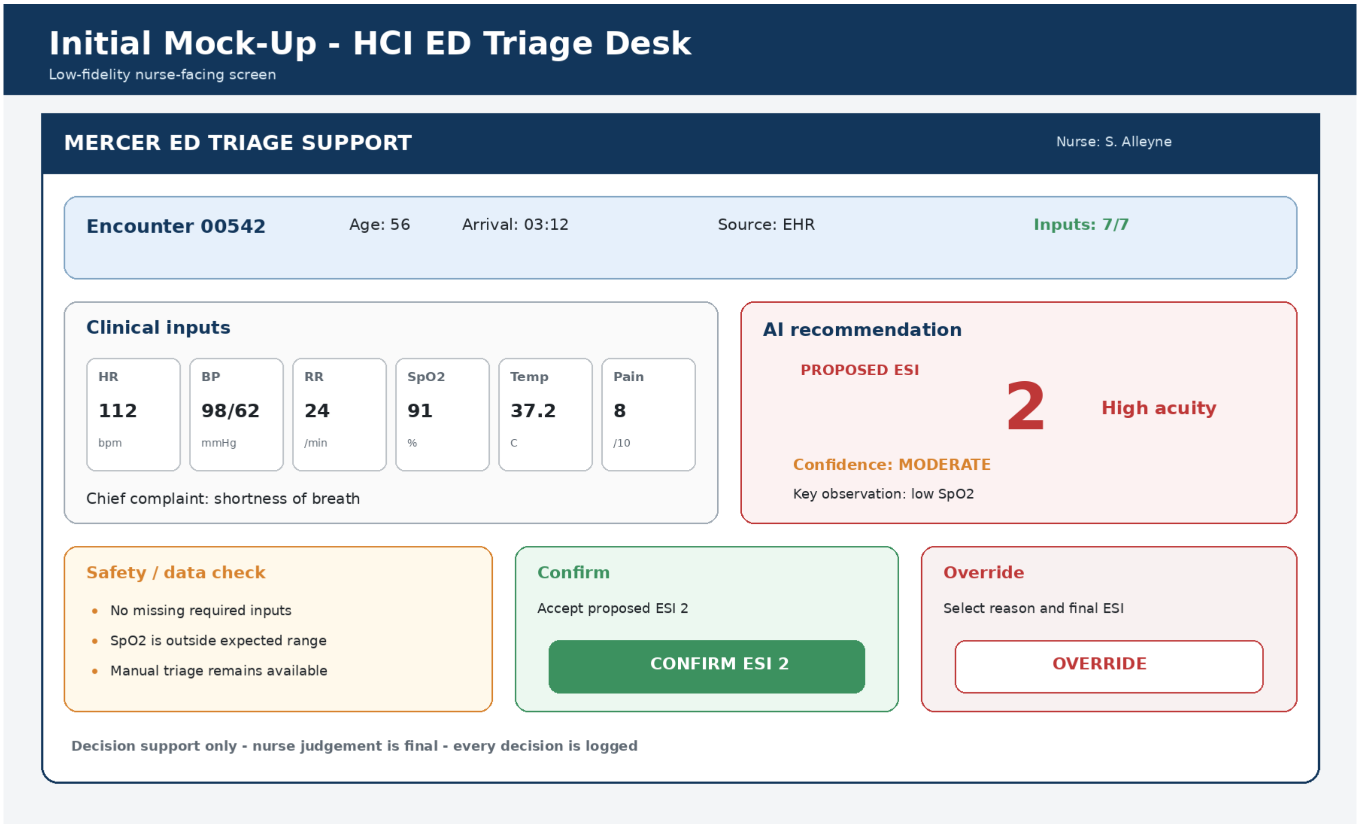
3. Proximity and physical safety

Concern	A robot or kiosk obstructs movement, startles a patient or creates a collision/infection-control hazard.
Context	The unit has limited space and mixed users, including weak, anxious or mobility-limited patients.
Mitigation	Stationary pilot, rounded form, marked clearance zone, emergency stop, low speed if movement is ever introduced, cleanable surfaces.
Residual risk	Human behaviour is unpredictable; on-site risk assessment and supervised testing remain necessary.

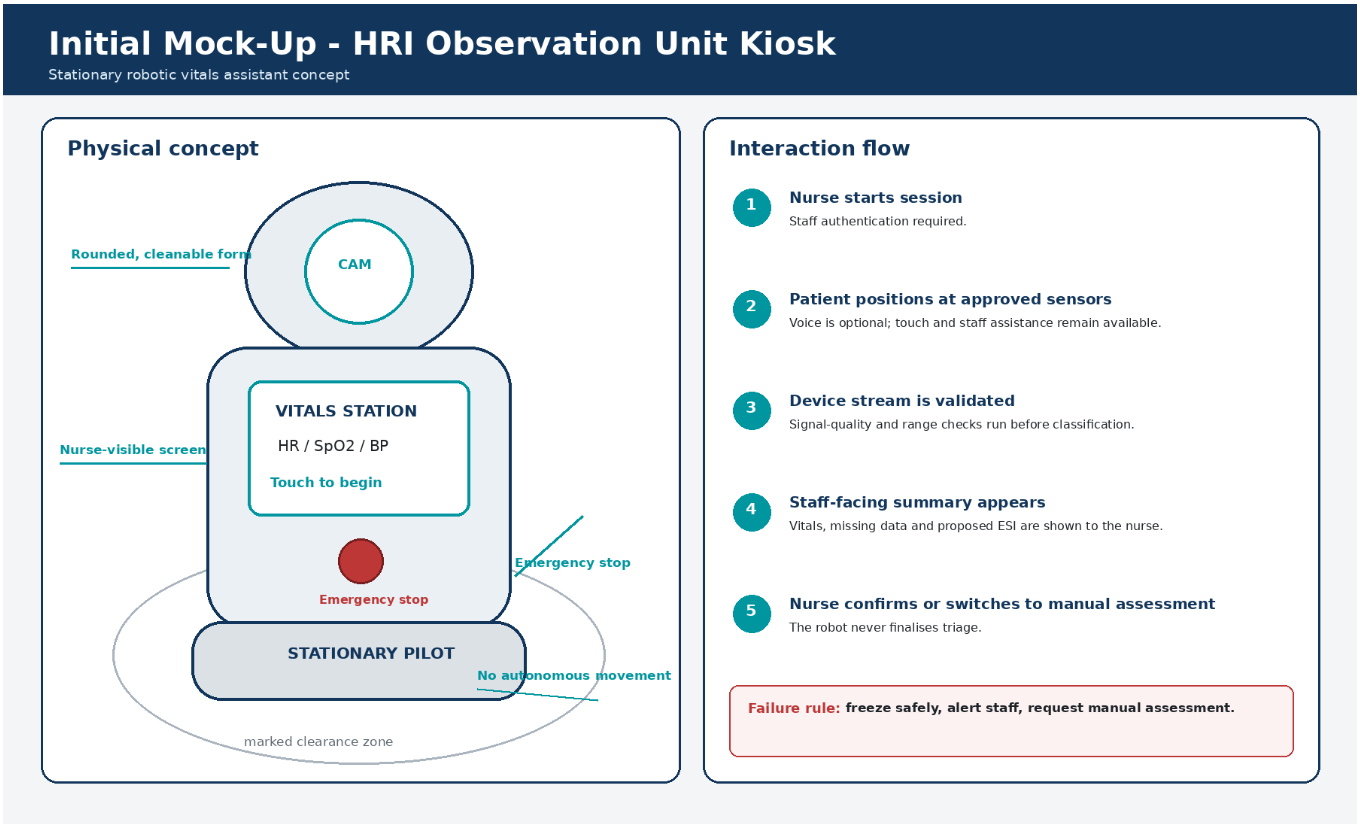
16. Failure modes and graceful degradation

Failure event	What the system does	What the human does	What is logged
Power loss	Screen goes dark; no recommendation is displayed.	Nurse immediately follows manual triage protocol.	Power event and last saved state logged when service returns.
Connectivity loss	EHR pull fails; offline banner appears.	Nurse selects manual entry within two actions.	Connection failure and manual mode logged.
Missing/invalid data	Inference is blocked and missing fields are highlighted.	Nurse rechecks values or performs manual assessment.	Missing-field event and source correction logged.
Low model confidence	Recommendation is labelled low confidence and not emphasised as final.	Nurse performs full manual review before deciding.	Confidence state and final decision logged.
User disagreement	Override control remains visible.	Nurse selects reason, enters optional note and confirms final ESI.	Recommendation, final ESI and override reason logged.
HRI sensor/proximity fault	Robot becomes stationary/non-interactive and displays staff alert.	Nurse removes patient from interaction and completes manual assessment.	Fault type, time and staff acknowledgement logged.

17. Initial Mock-Up - HCI ED Triage Desk



18. Initial Mock-Up - HRI Observation Unit Kiosk



19. Interim submission checklist

Deliverable	Status	Evidence
HCI problem-space canvas	**COMPLETE**	Included as PNG, PDF and in this document.
HRI problem-space canvas	**COMPLETE**	Included as PNG, PDF and in this document.
Preferred method canvas	**COMPLETE**	Preferred HCI implementation completed across Problem, Ethics, Guidelines, MVP, Environment and Form.
Initial mock-up sketches	**COMPLETE**	HCI dashboard and HRI kiosk concepts included.
System-requirement notes	**COMPLETE**	Inputs, outputs, human action, functional, non-functional and integration requirements included.
Safety comparison	**COMPLETE**	Three HCI and three HRI safety considerations included.

Recommended Google Drive upload

- Upload the combined PDF and DOCX.
- Upload the two individual problem-space canvas PDFs.
- Upload the preferred HCI co-design canvas PDF.
- Upload the initial mock-up PNG files or the combined mock-up PDF.
- Do not submit the tutorial screenshot PDF as your own deliverable unless the tutor asks for supporting material.

Source used

CariSurg MedTech Pathways, Week 9 tutorial screenshots: HCI/HRI definitions and design differences; Mercer environment constraints; inputs, outputs and human action; co-design canvas sections; mock-up examples; and safety-consideration format. Relevant tutorial pages: 13-16, 22-34, 35-55 and 85-95.